ASSESSMENT 2 — TAKE-HOME INDIVIDUAL PROJECT

Topic: Interactive 2D/3D Scene (Three.js or OpenGL)
Weighting: 15% of final grade
Format: Individual take-home project
Issued: Week 6
Due: Week 10

Learning Outcomes Assessed

- Design and implement an interactive graphics application using a modern graphics API/tool.
- Apply geometric transformations under user control.
- Configure a working camera with switchable projection.
- Communicate design decisions clearly in written form.

Task Description

Working individually, build an interactive 2D or 3D scene using Three.js (WebGL/WebGPU) or OpenGL/C++, incorporating all of the following:

- At least one object built from primitive geometry (e.g. cubes, spheres, custom vertex data) — not a single imported pre-made model.
- User-controlled transformations: keyboard and/or mouse input must let the user translate, rotate and/or scale at least one object in the scene in real time.
- A working camera with a control (key press, button or UI toggle) to switch between orthographic and perspective projection.
- Correct handling of objects that leave the view frustum (no visual glitches, crashes or runaway transformations).
- At least one original visual or interactive touch beyond the minimum requirements (e.g. colour change on interaction, a second controllable object, simple collision boundary).

Deliverables

- Source code, either as a zipped project folder or a link to a Git repository.
- A short report (1–2 pages) explaining your scene design, which transformations you implemented and how, and any challenges encountered.
- A 2–3 minute screen recording demonstrating all interactive features.
- A README with clear instructions to run the project (dependencies, commands, controls).

Submission Guidelines

- Submit via the LMS assignment link by 23:59 on the due date.
- Late submissions are penalised per the course outline's late-submission policy (Section 5.0.1).
- Work must be the student's own; cite any external code snippets or tutorials used.

Marking Rubric — Take-Home Individual Project (100 marks, scaled to 15%)

Criterion	Marks	Excellent	Satisfactory	Needs Improvement
Functionality & interactivity	25	All interactive controls work smoothly and	Most controls work with minor bugs that do not prevent core functionality.	Major features missing, broken, or the application crashes/misbehaves.

		reliably; no bugs or crashes during the demo.		
Correct use of transformations	20	Translation, rotation and/or scaling are correctly implemented and respond accurately to user input.	Transformations mostly work but with occasional inaccuracies (e.g. drift, incorrect pivot).	Transformations are missing, incorrect, or not connected to user input.
Camera & projection implementation	15	Camera switches cleanly between orthographic and perspective on demand; both views render correctly.	Camera switching works but with minor visual issues in one mode.	Camera switching is missing or does not function correctly.
Code quality & organisation	15	Code is well-structured, commented, and organised into sensible functions/modules.	Code is functional but only partially organised or commented.	Code is disorganised, uncommented, or very difficult to follow.
Report & documentation	15	Report clearly explains design decisions and challenges; README makes the project trivial to run.	Report and README cover the basics but lack some detail or clarity.	Report/README missing, very thin, or project cannot be run from the instructions given.
Creativity & visual design	10	Scene includes a genuinely original touch beyond the minimum brief; visually polished.	Scene meets the brief with a small original addition.	Scene is a bare-minimum implementation with no original elements.

Total: 100 marks